

Stoquify Customer Workflow Execution Report

Execution date: 2026-08-09

Workspace: E:\ohada saas\Focused projects\stoquify

Prompt source: attached Stoquify multidisciplinary customer-workflow execution prompt

Report classification: code-level remediation verified; production release evidence blocked

Statutory boundary: this report is not an OHADA filing certification, legal opinion, or Close & Assurance certificate.

1. Executive verdict

The customer workflow is materially stronger and the implemented code slice is verified. Customer routes now enforce the Sales entitlement, customer financial values use the tenant's authoritative currency rather than fixed XAF/USD labels, mixed-currency receivables are never collapsed into a false total, and the AR export now generates a real audited CSV rather than returning only an authorization decision.

The current workspace also contains a substantial pre-existing customer statement, settlement, receivable-document, external-access, delivery, and referral foundation. Its automated customer/AR regression suite is green.

Production readiness is still blocked. The repository does not contain a valid real-user pilot manifest, exact deployed revision, production PostgreSQL evidence, consented delivery evidence, external recipient evidence, or the complete statement-to-referral-conversion event sequence. The working tree is also heavily mixed with unrelated and staged changes, so no exact release revision can be certified from this workspace state.

Decision area	Verdict	Evidence
Customer master and route boundary	Ready at code level	Permission checks plus enforced Sales-module decision
Customer retention behavior	Ready at code level	Financial-history relations cause deactivation; regression coverage added
Customer exports	Ready at code level	Server-owned selection, redaction, fresh auth, watermark, audit, file download
AR history	Ready at code level	Bitemporal posted-document read model and currency-separated summaries
AR export	Ready at code level	Real CSV, server-derived row count, hashes, watermark, audit, download
Statements and settlements	Pilot-capable code foundation	Immutable snapshots, receivable evidence, idempotency, reversals, access/delivery models
Live statement/referral pilot	Blocked	0/16 production pilot checks ready
Exact release certification	Blocked	Dirty mixed scope, no isolated Git revision, no exact-revision security scan

2. Scope and method

The prompt was executed as an evidence-led audit plus the smallest coherent remediation slice. The audit covered:

- customer master lifecycle and retention;
- customer-to-sales/POS and receivable continuity;
- AR open items, allocations, aging, settlements, reversals, and statements;
- permissions, tenant isolation, module entitlement, fresh authentication, and auditability;
- exports, redaction, spreadsheet-injection safety, watermarks, hashes, and actual browser downloads;
- organization currency, mixed-currency aggregation, locale behavior, and residual localization gaps;
- data-model provenance, idempotency, immutable evidence, performance bounds, and release controls;

- current architecture graphs and the dirty-worktree/release boundary.

No migration, deployment, external message, database mutation, staging operation, commit, or push was performed. The repository was already heavily modified; all unrelated changes were treated as user-owned.

3. Architecture and ownership

3.1 Canonical ownership

Customer master, customer sales activity, and ordinary customer exports belong to the canonical Sales module. Immutable customer statements additionally cross into Accounting, which already has its own enforced entitlement on the statement route.

The runtime topology is:

```
Customer page/layout
-> RBAC customer permission
-> Sales module entitlement
-> protected customer action
-> tenant-scoped customer service
-> Sales/POS order
-> posted receivable document + lifecycle state
-> customer ledger / settlement allocation
-> AR open-item projection
-> immutable statement snapshot
-> consented external access/delivery/referral evidence
```

3.2 Graph findings

The existing action graph separates legacy customer actions into Community 3 (cohesion 0.22) and management/analytics actions into Community 13 (cohesion 0.41): [action graph report](#). This confirms that compatibility and management boundaries coexist and require source-level verification rather than assumptions from graph proximity.

The component graph places customer form/export helpers in a very broad Community 0 with cohesion 0.02: [component graph report](#). That graph is useful for navigation, but the low cohesion means source and tests remain authoritative for security and financial behavior.

The release-scope gate requires all architecture graphs to be regenerated only after an isolated customer/referral revision exists. Regenerating them in this mixed worktree would create misleading evidence.

4. Implemented and verified remediation

4.1 Enforced Sales entitlement for every customer route

The shared customer layout now performs:

1. `customers.read` permission enforcement;
2. authenticated tenant resolution;
3. an audited sales module access decision in enforce mode;
4. a `locked_module` route state when entitlement is absent.

Evidence: [customer layout/dashboard/customers/layout.tsx#L9](#)) and [layout boundary tests/dashboard/customers/__tests__/layout.test.tsx#L45](#)).

This closes the prior page-level gap where protected actions enforced Sales but the shared UI boundary only checked a customer permission.

4.2 Retention-safe customer removal

The current service checks sales orders, ledger entries, receivable documents, statement snapshots, and settlements before deciding whether a customer may be archived. Any financial or operational history forces deactivation rather than historical erasure.

Evidence: [customer lifecycle service](#) and [retention regression cases](#).

The new regression matrix explicitly covers receivable-document-only, statement-only, and settlement-only customers, plus the genuinely history-free archive path.

4.3 Authoritative customer currency

The customer management read model now loads and validates the organization currency. It returns a normalized ISO-style three-letter code with the customer dataset; absent or invalid currency fails closed.

Evidence: [customer management contract and query](#), [currency validation](#), and [dashboard formatting](#).

The customer order page independently resolves organization settings on the server, validates the currency, and passes it to the client formatter. The previous fixed USD formatter is gone.

Evidence: [customer order server page](#)/dashboard/customers/%5Bid%5D/orders/page.tsx#L25), [order client formatter](#)/dashboard/customers/%5Bid%5D/orders/CustomerOrdersClientPage.tsx#L58), and [fail-closed test](#)/dashboard/customers/%5Bid%5D/orders/ __tests__ /page.test.tsx#L96).

4.4 Currency-safe AR summaries

The AR result contract now exposes summariesByCurrency. For a single currency, the compatibility summary still contains monetary totals and an explicit currency. For mixed currencies, top-level monetary fields are null and mixedCurrency is true; no false combined amount is emitted.

Evidence: [AR result contract](#), [currency grouping](#), [AR workbench currency display](#), and [mixed-currency regression](#).

Rows, drawer details, allocations, KPI labels, and monetary columns now format using each item's actual currency.

4.5 Real, controlled AR export

The former AR export returned only a control decision. It now:

- reloads the authoritative server-side AR dataset;
- derives row count on the server and ignores the legacy client count;
- binds normalized filters, asOf, and recordedThrough;
- emits per-currency summary evidence;
- creates a durable watermark and content hash;
- requires verified fresh authentication and export permission;
- records the allow/deny audit decision inside the transaction;
- escapes spreadsheet formula prefixes;
- returns an actual CSV that the client downloads.

Evidence: [AR export service](#), [audit before release](#), [server action](#), [download hook](#), and [export regressions](#).

4.6 Existing customer export controls verified

The current customer export path already:

- derives tenant scope from protected action context;
- checks Sales entitlement and customer export/read permissions;
- selects rows on the server;
- defaults to redacting email, phone, address, tax ID, credit limit, current balance, and notes;
- prevents CSV formula execution;
- applies fresh authentication, watermark, filter hash, content hash, and audit evidence;
- downloads the generated file only after success.

Evidence: [customer action controls](#), [redacted fields](#), [CSV injection control](#), [audit decision](#), and [download hook](#).

5. End-to-end workflow audit

Domain	Current evidence	Classification	Residual concern
Customer create/read/update	Protected management actions, tenant-derived organization ID, validated schemas	Ready at code level	No live browser evidence in this run
Search/filter/analytics	Server read model, bounded 500-row management surface, filtered exports	Ready at code level	Large-tenant query plan not benchmarked
Customer archive/delete	Soft lifecycle behavior with broad history check	Ready at code level	Concurrent lifecycle mutation should be stress-tested in PostgreSQL
Sales/POS linkage	Customer orders and posted receivable-document foundation	Ready at code level	SalesOrder remains organization-base-currency, not a proven per-order FX model
Receivable documents	Immutable document identity, hashes, append-only lifecycle state	Ready at code level	Requires migration deployment evidence on target DB
AR open items	Posted state, allocations, aging, asOf and recordedThrough boundaries	Ready at code level	Live data reconciliation not run
Settlement collection	Idempotency, allocation, ledger/accounting links	Pilot-capable	Provider/bank operational evidence not supplied
Settlement reversal	Compensating reversal foundation and protected action tests	Pilot-capable	Production maker-checker evidence not supplied
Statements	Immutable, currency-specific snapshots with source hashes	Pilot-capable	Production generation and delivery evidence absent
External statement access	Token hashes, redacted access logs, disputes/promises-to-pay	Pilot-capable	Real recipient cohort and abuse testing absent
Statement delivery/referral	Consent-bound delivery and referral event models	Code foundation present	Live end-to-end event sequence is blocked
Customer export	Controlled, redacted, watermarked, audited actual file	Ready at code level	Retention policy for downloaded local files is outside application control
AR export	Controlled, currency-explicit, bitemporal actual file	Ready at code level	No production-volume export test
Tenant isolation	Protected action context and tenant-scoped service queries	Ready at code level	No live cross-tenant penetration test in this run
RBAC and module access	Customer permissions plus Sales/Accounting entitlements	Ready at code level	Exact production role assignment evidence absent
Localization	Customer management supports English/French copy	Partial	Customer orders client remains substantially English-only
Accessibility/responsiveness	Existing component tests and dashboard primitives	Not certified	No authenticated desktop/mobile browser or assistive-technology run

Domain	Current evidence	Classification	Residual concern
Observability	Audit logs, evidence hashes, business-event links	Pilot-capable	No production alert/dashboard proof supplied

6. Security, privacy, and financial-control assessment

Controls verified

- Server actions do not trust a caller-supplied organization ID as authority.
- Customer and AR exports require specific permissions and fresh authentication.
- Controlled exports are watermarked, hashed, audited, and spreadsheet-safe.
- AR summaries preserve currency identity and do not add unlike currencies.
- Statements are currency-specific and bind asOf, recordedThrough, source hashes, and idempotency evidence.
- Historical customer evidence prevents destructive lifecycle behavior.
- Public statement evidence uses token hashes and redacted access logs rather than raw secrets.

Residual risks

1. P0 — production pilot evidence missing. The referral pilot gate reported 0/16 ready checks.
2. P0 — exact release cannot be isolated. The worktree contains unrelated staged and unstaged changes, mixed shared files, and unclassified paths.
3. P1 — full transaction-level multi-currency is not proven. The remediation prevents mislabeled and mixed aggregation, but ordinary sales still inherit organization currency; a true per-order FX/rate model would require explicit product and accounting design.
4. P1 — order-history localization is incomplete. The customer order client contains substantial English-only copy.
5. P1 — contact toasts expose full email/phone values. [Customer quick actions](#) place complete contact data in local notifications before launching mailto: or tel:; redaction is preferable.
6. P1 — production-volume performance is unmeasured. The management page is bounded to 500 rows ([service limit](#)), but PostgreSQL query plans and export memory behavior were not benchmarked.
7. P2 — browser and accessibility evidence is absent. No authenticated runtime, mobile viewport, keyboard-only, or screen-reader session was available.

7. Verification evidence

Automated tests

Command/scope	Result	
Baseline customer actions/export and AR service	3 suites, 20 tests passed	
Focused remediation set	5 suites, 30 tests passed	
Customer layout exact-path test	1 suite, 2 tests passed	
Broad `customer`	ar-open-item` regression	34 suites, 197 tests passed
Customer order currency boundary exact-path test	1 suite, 4 tests passed	

The broad run includes customer master, lifecycle, receivable documents, ledger integrity, settlements, reversals, statements, access tokens, recipient actions, delivery, referral gates, API routes, components, and AR services.

Static verification

Check	Result
TypeScript tsc --noEmit	Passed after final changes
Focused ESLint on changed customer/AR files	Passed
git diff --check on scoped files	Passed; only existing CRLF normalization warnings
Report trust/export gate	Ready, 35/35 checks, 0 blockers
Hard-delete gate	0 active unsafe findings

Release and operational gates

The real-user customer referral pilot gate is blocked:

- 0/16 checks ready;
- placeholder pilot ID;
- no exact release revision;
- no PostgreSQL target;
- no read-only production evidence query;
- no immutable statement, sent delivery, external view, recipient response, referral impression/click/conversion, or referred-organization activation evidence.

The release-scope report is also blocked because:

- unclassified paths remain;
- unrelated user index changes are present;
- four shared files require hunk isolation;
- architecture graphs must be regenerated on an isolated revision;
- no exact referral Git revision exists;
- no security diff scan exists for that exact revision.

These are legitimate external/release blockers, not code-test failures.

8. Dependency-ordered completion plan

P0 — isolate and bind the release

1. Create a dedicated customer/referral release revision without unrelated staged or working-tree changes.
2. Classify the remaining unclassified paths and isolate the four shared-file hunks.
3. Regenerate architecture graphs on that exact revision.
4. Run a security diff scan and bind its digest to the exact revision.

P0 — execute the production pilot

1. Deploy the existing migrations to an approved PostgreSQL pilot target.
2. Supply a non-placeholder pilot manifest, tenant consent, exact revision, and evidence references.
3. Execute the statement-to-delivery-to-view-to-response-to-referral-to-activation sequence with real, consented, non-seed data.
4. Re-run the read-only pilot evidence gate until all 16 checks are ready.

P1 — finish product and control hardening

1. Decide whether Stoquify supports only organization base currency or true per-order currency/FX; if the latter, add explicit currency/rate identity and posting rules.
2. Localize the customer order workbench fully in English and French.
3. Redact contact data in client notifications.
4. Run PostgreSQL query-plan and memory tests for 500-row management reads and large controlled exports.
5. Add authenticated browser tests for create/edit/archive, order navigation, exports, statement generation, module lock, empty/error states, and mobile layouts.

P2 — release assurance

1. Run keyboard, focus, screen-reader, contrast, and responsive checks.
2. Capture operator, accountant, and customer-recipient evidence.
3. Verify production audit-log ingestion, export alerts, delivery dead-letter handling, and settlement exception monitoring.
4. Route immutable statement and close evidence through Stoquify Close & Assurance before making certification claims.

9. Files changed by this execution

- app/[locale]/(dashboard)/dashboard/customers/layout.tsx
- app/[locale]/(dashboard)/dashboard/customers/__tests__/layout.test.tsx
- app/[locale]/(dashboard)/dashboard/customers/[id]/orders/page.tsx
- app/[locale]/(dashboard)/dashboard/customers/[id]/orders/CustomerOrdersClientPage.tsx
- app/[locale]/(dashboard)/dashboard/customers/[id]/orders/__tests__/page.test.tsx
- actions/finance/ar-history.actions.ts
- components/customers/CustomerManagementDashboard.tsx
- components/finance/AROpenItemsHistoryWorkbench.tsx
- hooks/useAROpenItemsHistoryWorkbench.ts
- messages/en.json
- messages/fr.json
- services/accounting/ar-open-item.service.ts
- services/accounting/ar-open-item-export.service.ts
- services/accounting/__tests__/ar-open-item.service.test.ts
- services/accounting/__tests__/ar-open-item-export.service.test.ts
- services/customer/customer.service.ts
- services/customer/__tests__/customer-legacy.service.test.ts

10. Final conclusion

The executed slice closes the most material verified customer-workflow defects that were safe to resolve in the current workspace: missing page entitlement enforcement, invented customer currencies, unsafe mixed-currency totals, and a non-file AR export. The customer/AR automated regression and static control gates pass.

The honest release verdict remains blocked pending isolated revision and live pilot evidence. Nothing in this report should be interpreted as production deployment approval, statutory certification, or evidence that external delivery and referral conversion occurred.